Robustifying the Psyche reference trajectory

Multiphase stochastic optimisation, launch covariance,
and an event-aware scaled B-plane safety constraint

Prepared for Lukas Messina • 25 October 2026

Research and implementation proposal. No optimisation implementation has been changed.

1 Adopted revisions and remaining decisions

Retain the existing JAX workflow: a deterministic energy reference, a discrete TVLQR gain seed, nonlinear moment propagation, and simultaneous optimisation of the mean trajectory, feedback gains and covariance. The revised problem starts at the **SPICE-reconstructed Earth SOI exit**; the Earth-centred launch/departure phase is removed. The retained phases are Sun, Mars and Sun, with **50 / 116 / 50 arcs**, 216 in total. Preserve the phase-normalised mesh coordinates of the corresponding three phases of the saved energy optimum. Start with four uniform XMDS2 RK9 substeps per arc for the JAX deterministic and stochastic maps and verify accuracy before trusting a solve.

The launch-parameter bounds on printed page 12 of the thesis are now interpreted, by the user's instruction, as independent Gaussian 3σ limits using Eq. (5.2). The initial position standard deviation is **2000 km per Cartesian axis**. Use the asymptotic Cartesian velocity covariance from Section 5 as an approximate covariance at Earth SOI exit, centred on the actual SPICE state. The Mars flyby risk is fixed at $\epsilon_M = 0.001$, corresponding to a 99.9% safety target within the adopted chance-constraint approximation.

For Mars safety, evaluate the osculating hyperbolic B-plane at nominal closest approach, differentiate its moving axes and gravitational-focusing scale, and include the shift to each perturbed trajectory's own closest approach. This follows Farnocchia et al. [1], especially Sections 4–7, and the chance-constraint construction of Anibha et al. [2]. Nuccio et al. [3] additionally motivate a separate loss-of-maneuverability path constraint. These two safety requirements remain distinct.

The following choices remain for review:

1. **Mars minimum altitude.** Select h_{safe}; the thesis leaves it TBD. The risk is now fixed at 0.001 and is no longer an open choice. No altitude is invented here.
2. **Safety scope.** Recommended first implementation: safety of the actual controlled Mars flyby. Add the passive loss-of-thrust scenario of Section 13 only if that additional failure requirement is desired.
3. **Nonuniform-mesh cost and arrival covariance.** Recommend the time-weighted extension in Section 8, which equals the current cost on a uniform mesh. Recommend initially retaining Earth–Mars's absolute terminal benchmark of 2000 km and 2 m/s per axis, subject to approval; it is not a Psyche navigation requirement.

Fix the nominal/mean Cartesian state and epoch at Earth SOI exit, Mars SOI entry and Psyche arrival. Only Mars SOI exit remains free. Individual random trajectories are dispersed and are not snapped to these states. In particular, there is no covariance reset at Mars entry. The initial Gaussian is defined at the common fixed Earth-exit epoch; its samples are not individually forced onto the SOI surface.

This revision changes the proposed stochastic problem only. The existing four-phase deterministic implementation and its saved results have not been modified or rerun. The former finite-TIP conversion and nonlinear-TIP study (Sections 5.3–5.4 of the first proposal) have been removed.

2 Evidence and the deterministic reference

The inspected public energy result is

> `output/psyche_mission_deterministic_traj_opt/energy_optimal/psyche.npz` and its accompanying `psyche.json`.

The source archive is `shooting_05.npz`, under the run directory

> `.cache/deterministic_runs/20261024_185218_432429/psyche_energy/`

It reports IPOPT success, energy 609809.272913296 N^2 s, propellant 814.520099492529 kg, and a final 296-arc mesh for the original four-phase problem. It is the parent reference, not a converged solution of the revised three-phase problem. The fuel optimum is not used to determine the mesh.

Retained parent phase	Arcs	Parent duration (days)	Parent arc durations (s)
First solar cruise	50	941.940380002	1627672.977
Mars flyby	116	2.967385416	178.043–2848.690
Final solar cruise	50	1172.204612743	2025569.571

The omitted parent Earth-departure phase had 80 arcs. Retain only `p1_mesh`, `p2_mesh` and `p3_mesh` and their native state/control arrays; reindex these as the three active phases. The new fixed start is 29.15985835 s later than the parent's optimised Earth exit, and its Cartesian state also differs. The first cruise now lasts 81383619.67227745 s (1627672.39344555 s per initial retained arc). The parent nodes and controls are a seed requiring a new energy restoration; simply deleting the first phase does not produce a converged solution with the new boundary. The very different durations reinforce why a per-arc unweighted cost cannot be carried over without discussion. The common plotting arrays are Sun-centred; using them to recover planet-relative states by subtracting large translations would unnecessarily lose precision.

The repository and attached thesis copies have identical SHA256 hashes. The document is titled *GPU-Accelerated Robust Trajectory Optimisation for Low-Thrust Space Missions*; its supplied version contains unfinished requirements. Exact file hashes, the nonuniform meshes and numerical checks accompany this proposal in `review_manifest.json` and `numerical_review.json`. The equations from the attached papers were checked against their rendered pages, not only PDF text extraction.

2.1 Fixed events and model provenance

All origins use geometric, equatorial J2000 orientation. Epochs are TDB seconds past J2000; dynamics are in SI before numerical scaling. The fixed Cartesian state vectors are the existing values in the reconstruction JSON files, reproduced in Appendix **??**.

Event	Epoch (TDB seconds)	Boundary/source
Earth SOI exit	750633519.6868194	Sun-relative spacecraft state; 2023-10-15 09:17:30.504454 UTC; reconstructed launch SPK.
Mars SOI entry	832017139.3590969	Mars-centre-relative spacecraft state; 2026-05-14 07:51:10.173817 UTC; reconstructed Mars SPK.
Psyche arrival	933552000.0	Asteroid position and velocity; 2029-08-01 12:00 TDB; selected asteroid prediction.

The arrival is the previously selected mathematical rendezvous with the asteroid, not a measured spacecraft capture state. The fixed-semimajor-axis Laplace radii are

$$R_{\mathrm{SOI,p}} = a_p(\mu_p/\mu_\odot)^{2/5}, \qquad R_{\mathrm{SOI,E}} = 924646789.2005551 \text{ m}, \quad R_{\mathrm{SOI,M}} = 577239186.7537631 \text{ m}. \tag{2.1}$$

Here $a_E = 1$ au and $a_M = 1.52371034$ au. Earth exit is now a fixed reconstructed input. Only the Mars exit epoch remains an optimisation variable, with $\|\bar{r}\| = R_{\mathrm{SOI,p}}$ and $\bar{r}^{\mathsf{T}}\bar{v}/\|\bar{r}\| \geq 0$. The implemented nonnegative inequality is not a strict > 0 constraint. The Mars-centred duration remains between 0.05 and 10 days; the Earth duration variable and its bounds are removed. No inside/outside-SOI path inequalities are added.

The Earth-exit event was re-refined with TudatPy/SPICE using the pinned `psyche_rec_231013-231018_v1.bsp`. Its spacecraft coverage is ET 750482577.1393577–750916869.182383 s, which contains the crossing. The epoch matches the retained event record exactly; the radius residual is 2.51×10^{-4} m and the outward radial speed is 5898.7386 m/s. A raw CSPICE query reproduces the Tudat state after unit conversion. This checks the interface and reconstruction, not independent flight accuracy. The segment-origin change occurs 0.374792 s later and is not used as the boundary.

Spacecraft parameters retain $m_0 = 2747$ kg at the new initial epoch, $m_{\min} = 1662$ kg, $T_{\max} = 0.2793$ N, $I_{\mathrm{sp}} = 1794$ s and $g_0 = 9.80665\ \mathrm{m\,s^{-2}}$. Surface radii are Earth 6378136.6 m, Mars 3396190 m and Sun 695700000 m. These are inherited model choices. With the Earth phase removed, the active central-body surfaces are Mars and Sun. Retaining 2747 kg at Earth exit is an explicit initial-mass assumption: SPICE supplies no spacecraft mass and the omitted departure propellant is not modelled. The parent energy solution had 2746.500203 kg at its own optimised exit; that value is not silently substituted. Planetary ephemerides are treated as deterministic; no planet-orbit covariance or timing error is invented. The existing cache uses cubic splines on a 5400 s grid, with checked coverage and derivatives. Its full provenance remains in `psyche_mission_events_reconstruction/README.md`.

3 Three-phase dynamics and boundary sensitivities

Let $x = (r^{\mathsf{T}}, v^{\mathsf{T}}, m)^{\mathsf{T}} \in \mathbb{R}^7$ and $c = T/T_{\max} \in \mathbb{R}^3$. In each native phase,

$$\begin{aligned} dr &= v\,dt, \\ dv &= \left(-\frac{\mu_p r}{\|r\|^3} + \frac{T_{\max} c}{m}\right) dt + \sigma_a\,dW_t, \\ dm &= -\frac{T_{\max}}{I_{\mathrm{sp}} g_0} q_\eta(c)\,dt, \qquad q_\eta(c) = \sqrt{c^{\mathsf{T}}c + \eta^2}, \quad \eta = 10^{-5}. \end{aligned} \tag{3.1}$$

The central bodies are Sun, Mars and Sun; no Earth-centred departure propagation is included. The smoothed norm is the *current stochastic script's* formulation, unlike the deterministic archive's exact throttle mass flow. Retaining it is an explicit methodological choice. It adds at most $T_{\max}\eta(t_f - t_0)/(I_{\mathrm{sp}} g_0) \simeq 0.0290$ kg of fuel overestimate for a given control history. It does not introduce an epigraph slack.

Let the saved phase fractions be $s_{p,k}$. For variable phase endpoints $t_{p,a}, t_{p,b}$,

$$t_{p,k} = t_{p,a} + s_{p,k}(t_{p,b} - t_{p,a}), \quad \Delta t_{p,k} = (s_{p,k+1} - s_{p,k})(t_{p,b} - t_{p,a}). \tag{3.2}$$

Only one independent duration variable is needed: the Mars flyby duration. The first cruise duration is fixed by Earth SOI exit and Mars SOI entry. The final cruise duration follows from the variable Mars exit and the fixed arrival epoch. The mesh topology stays fixed; Mars exit is not frozen to the deterministic optimum.

At a change from body p to the Sun,

$$x_\odot = \mathcal{R}_p(t, x_p) = \begin{bmatrix} r_p + r_{p/\odot}(t) \\ v_p + v_{p/\odot}(t) \\ m_p \end{bmatrix}. \tag{3.3}$$

Reverse the signs for the reverse translation. At a fixed deterministic epoch, $D_x\mathcal{R} = I_7$, so physical covariance is unchanged by this translation. In numerically scaled coordinates, covariance must be transformed with the corresponding scale matrices; it cannot simply be copied as a dimensionless array. The time derivative is

$$\mathcal{R}_t = \begin{bmatrix} \dot{r}_{p/\odot}(t) \\ \dot{v}_{p/\odot}(t) \\ 0 \end{bmatrix}. \tag{3.4}$$

Use derivatives of the *actual stored splines*; fitted position and velocity are not exactly kinematically identical. Duration and spline derivatives enter the shooting residuals, gains and covariance sensitivities. No SPICE calls occur in optimisation.

The baseline *does not use a saltation-matrix update* at Mars SOI entry or exit. All stochastic samples use the nominal phase schedule: Mars entry remains fixed, while the nominal Mars exit epoch is optimised. A free design epoch is not itself a random epoch; covariance is transformed with the coordinate and numerical-scale Jacobians at these scheduled boundaries.

A planned extension will include third-body gravity in every phase: solar perturbations in the Mars-centred phase and Martian perturbations in the heliocentric phases, with any additional perturbing bodies treated consistently across phases. Third-body terms include the acceleration of the propagation origin, as in Tudat's differential formulation [6]. With consistent origin ephemerides and the same physical thrust/control policy across the boundary, the two phase descriptions should represent the same physical dynamics. In the exact consistent limit,

$$\Delta f \equiv f^+ - D_x\mathcal{R} f^- - \mathcal{R}_t = 0, \qquad \Xi = D_x\mathcal{R}.$$

For that future model, the additional state/covariance error associated with state-dispersion-induced timing deviations at Mars SOI entry and exit is expected to arise primarily from residual numerical cancellation, ephemeris interpolation, and consistency between the prescribed planetary motion and the implemented force models. Accordingly, no saltation correction is included in the selected baseline. This expectation concerns the additional effect of changing phase at a displaced epoch; it does not imply that crossing-time dispersion or closest-approach uncertainty vanishes. The third-body extension is future work: the central-body equations above have not yet been replaced, and the size of the residual has not yet been numerically verified. In the monte carlo validation the dynamical model switch will happen at each single monte carlo trajectory exact soi crossing time, NOT the crossing time given by the optimizer.

4 Uncertainties inherited from Earth–Mars

The source of these numbers is `default_options("earth_mars")` and the common constants in `stochastic_traj_opt_jax.py`. They are inherited assumptions, not Psyche flight covariance products.

Quantity	Value	Interpretation
Initial position	2000 km/axis	User-requested 1σ, replacing Earth–Mars's 30000 km.
Initial velocity	Section 5	Anisotropic Cartesian asymptotic covariance, approximated at Earth exit.
Initial mass	0.001 kg	Current Earth–Mars 1σ; thesis says 0.01 kg.
Navigation position	200 km/axis	Independent observation error at each control update.
Navigation velocity	0.1 m/s/axis	Same update model.
Acceleration diffusion	10^{-10} km s$^{-3/2}$	10^{-7} m s$^{-3/2}$, not an acceleration standard deviation in m s^{-2}.
Gates magnitude	0.008	Proportional standard deviation.
Gates pointing	0.0095993 rad	Proportional small-angle standard deviation.
Control violation risk	0.01	Inherited per-arc thrust risk, unchanged.
Mars flyby risk	0.001	User-selected encounter risk (99.9% safety target).

With a local numerical scale $D = \mathrm{diag}(d_r I_3, d_v I_3, d_m)$, use $\widetilde{P} = D^{-1} P D^{-\mathsf{T}}$ and the same transformation for navigation covariance. For the existing dimensionless dynamics, $\sigma_{a,\mathrm{nd}} = \sigma_{a,\mathrm{SI}} t_*^{3/2}/\ell_*$.

5 Gaussian launch dispersion at the Earth SOI-exit start

5.1 What the thesis specifies

Thesis Section 2.5, printed page 12 (PDF page 30), gives

$$\begin{aligned} C_0 &= 34.0 \text{ km}^2\,\text{s}^{-2}, & \alpha_0 &= 1.9845214 \text{ rad}, & \delta_0 &= 0.4888475 \text{ rad}, \\ |\Delta C| &\le 0.25 \text{ km}^2\,\text{s}^{-2}, & |\Delta\alpha| &\le 0.0087266 \text{ rad}, & |\Delta\delta| &\le 0.0043633 \text{ rad}. \end{aligned} \tag{5.1}$$

The source states delivery bounds. For this study, the user explicitly adopts the Gaussian interpretation that each half-width is three standard deviations. This is an agreed modelling interpretation, not a newly established confidence statement from the launch provider. Initial Cartesian position dispersion at Earth SOI exit is 2000 km per axis.

For the adopted independent Gaussian interpretation, set $q = (C, \alpha, \delta)^{\mathsf{T}}$ and

$$P_q = \mathrm{diag}\Big[(0.25/3)^2, (0.0087266/3)^2, (0.0043633/3)^2\Big]. \tag{5.2}$$

Use $\Delta q \sim \mathcal{N}(0, P_q)$. The Gaussian has unbounded tails; these are 3σ limits rather than absolute bounds. Independence is the adopted assumption because no launch correlation matrix is supplied. No uniform-distribution alternative is retained in the active proposal.

5.2 Cartesian velocity covariance and initial Gaussian state

Using thesis Eq. (3.15),

$$v_\infty(q) = \sqrt{C}\, s(\alpha, \delta), \qquad s = \begin{bmatrix} \cos\delta\cos\alpha \\ \cos\delta\sin\alpha \\ \sin\delta \end{bmatrix}. \tag{5.3}$$

The Jacobian, with column order (C, α, δ), is

$$J_\infty = \left[\frac{s}{2\sqrt{C}}, \quad \sqrt{C} \begin{bmatrix} -\cos\delta\sin\alpha \\ \cos\delta\cos\alpha \\ 0 \end{bmatrix}, \quad \sqrt{C} \begin{bmatrix} -\sin\delta\cos\alpha \\ -\sin\delta\sin\alpha \\ \cos\delta \end{bmatrix} \right], \quad P_{v_\infty} = J_\infty P_q J_\infty^{\mathsf{T}}. \tag{5.4}$$

Evaluate J_∞ at the thesis nominal C_0, α_0, δ_0.

For the revised boundary, adopt

$$\bar{x}_0 = \begin{bmatrix} r_{\mathrm{SPICE},\odot}(t_0) \\ v_{\mathrm{SPICE},\odot}(t_0) \\ 2747\ \mathrm{kg} \end{bmatrix}, \quad P_{x,0} = \mathrm{blkdiag}\Big[(2\times 10^6\ \mathrm{m})^2 I_3,\ P_{v_\infty},\ (10^{-3}\ \mathrm{kg})^2\Big]. \tag{5.5}$$

The initial Cartesian Gaussian is defined directly: its mean and its central nominal state are both the SPICE state in Eq. (5.5). There is no nonlinear TIP-to-Cartesian mean transformation to decide in this revised model. Position and velocity errors are assumed independent at this new initial epoch; retain the off-diagonal entries inside the velocity block. Earth ephemerides are deterministic, so translating this error covariance into Sun-centred J2000 does not change it. The velocity uncertainty is added to the SPICE mean, not used to replace that mean by the asymptotic velocity vector.

The approximation is $\delta v_{\mathrm{exit}} \approx \delta v_\infty$: starting at the SOI is not mathematically the same as being at infinity. For the reconstructed crossing, the Earth-relative speed is 5899.449 m/s, whereas its osculating hyperbolic-excess speed is 5825.919 m/s, a difference of 73.530 m/s; $2\mu_E/R_{\mathrm{SOI,E}} = 0.862168\ \mathrm{km}^2/\mathrm{s}^2$. That speed difference is not a covariance error estimate. It establishes why the adopted covariance transfer must be labelled approximate even though the mean boundary comes directly from SPICE. This proposal does not reconstruct an exact launch-to-SOI covariance or reintroduce a TIP propagation.

6 Energy restoration and discrete TVLQR

The saved energy optimum is the reference input. The existing stochastic workflow restores dynamic consistency by solving a deterministic shooting problem before computing gains. For Psyche, do so on the retained 216-arc mesh, with the revised fixed Earth-exit boundary, the fixed Mars-entry and arrival boundaries, one free Mars exit epoch, and frame links.

$$J_E = \frac{1}{2} T_{\max}^2 \sum_k \Delta t_k \left\| c_k \right\|^2, \qquad x_{k+1} = \Phi_k(x_k, c_k; \Delta t_k). \tag{6.1}$$

The current code minimises a dimensionless multiple of this expression using $q_\eta(c)^2$; its extra $\eta^2 \sum \Delta t$ is constant because the total mission duration is fixed. Report physical energy without that artificial constant. The smoothed mass flow can still shift the restored solution slightly.

Differentiate the actual discrete map, including origin changes:

$$A_k = D_x \Phi_k, \quad B_k = D_c \Phi_k, \qquad \widetilde{A}_k = D_{k+1}^{-1} A_k D_k, \quad \widetilde{B}_k = D_{k+1}^{-1} B_k. \tag{6.2}$$

Here B_k is with respect to normalised thrust; if differentiated with respect to newtons, multiply by $T_{\max}$. The current TVLQR seed uses only the upper six state channels: $A_{6,k} = \widetilde{A}_k[1{:}6, 1{:}6]$, $B_{6,k} = \widetilde{B}_k[1{:}6, :]$. Mass remains in the seven-state nonlinear stochastic propagation, but is neither observed by feedback nor used in this Riccati seed.

$$\begin{aligned} K_k &= -\left(R_k + B_{6,k}^{\mathsf{T}} S_{k+1} B_{6,k}\right)^{-1} B_{6,k}^{\mathsf{T}} S_{k+1} A_{6,k}, \\ S_k &= Q_k + K_k^{\mathsf{T}} R_k K_k + (A_{6,k} + B_{6,k} K_k)^{\mathsf{T}} S_{k+1} (A_{6,k} + B_{6,k} K_k). \end{aligned} \tag{6.3}$$

For the present Earth–Mars settings, $Q = I_6/9$ and $R = 1000 I_3$, not I_3; S_N follows the chosen terminal covariance scales. With time weights approved below, use $Q_k = w_k Q$, $R_k = w_k R$ and document this corresponding seed extension. TVLQR is an initial guess, not a claim that its gains solve the nonlinear stochastic problem.

7 Feedback, execution error and nonlinear moment propagation

At each node observe the six-dimensional deviation with independent noise $\nu_k \sim \mathcal{N}(0, R_{\mathrm{nav,k}})$ and command

$$c_k^{\mathrm{cmd}} = s_k + K_k\big(D_{6,k}^{-1}(x_{6,k} - \bar{x}_{6,k}) + \widetilde{\nu}_k\big). \tag{7.1}$$

The command and its execution error are held constant during the whole arc; feedback is not continuously recomputed at every RK stage. For each commanded vector, the proportional Gates model is

$$\begin{aligned} c_k^{\mathrm{exe}} &= c_k^{\mathrm{cmd}} + e_{G,k}, \\ e_{G,k} \mid c_k^{\mathrm{cmd}} &\sim \mathcal{N}(0, Q_G(c_k^{\mathrm{cmd}})), \\ Q_G(c) &= \sigma_\theta^2(c^{\mathsf{T}}c)I_3 + (\sigma_{\mathrm{mag}}^2 - \sigma_\theta^2)cc^{\mathsf{T}}. \end{aligned} \tag{7.2}$$

Mass flow and acceleration both use the executed vector through Eq. (3.1); mass flow uses its smoothed norm. The current six-point inner transform adds a $10^{-12}I_3$ numerical jitter to Q_G, so at zero command it has a tiny numerical variance rather than exactly zero. This must remain visible in the numerical audit.

For $M = 4$ integration substeps, augment the seven-state uncertainty with six navigation variables and $3M$ independent standard-normal Wiener variables: $d = 7 + 6 + 3M = 25$. At the current $\kappa = 0$, the outer transform has

$$z^{(0)} = \bar{z}, \quad z^{(\pm i)} = \bar{z} \pm \sqrt{d}\, L_z e_i, \quad W_0 = 0, \quad W_{\pm i} = \frac{1}{2d}, \quad L_z L_z^{\mathsf{T}} = P_z. \tag{7.3}$$

The source uses its regularised Cholesky factor; the mathematical form here shows the intended covariance. For each outer command $c^{(i)}$, use six conditional execution points

$$c^{(i,\pm j)} = c^{(i)} \pm \sqrt{3}\, L_G(c^{(i)})e_j, \quad V_{\pm j} = 1/6, \quad j = 1, 2, 3. \tag{7.4}$$

Each substep receives $\Delta W_\ell = \sqrt{\Delta t_k / M}\, \xi_\ell$. Propagate all points with the same physical arc duration and construct

$$\begin{aligned} \bar{x}_{k+1} &= \sum_{i,j} W_i V_j X_{k+1}^{(i,j)}, \\ P_{k+1} &= \sum_{i,j} W_i V_j (X_{k+1}^{(i,j)} - \bar{x}_{k+1})(X_{k+1}^{(i,j)} - \bar{x}_{k+1})^{\mathsf{T}}, \\ C_{c,k} &= \sum_{i,j} W_i V_j (c_k^{(i,j)} - \bar{c}_k)(c_k^{(i,j)} - \bar{c}_k)^{\mathsf{T}}. \end{aligned} \tag{7.5}$$

Covariance defects enforce equality with these moment maps. The executed-control covariance includes execution error; it is not only KPK^{T}. Ignoring numerical jitter, a useful independent check is

$$\begin{aligned} C_{\mathrm{cmd}} &= K_k(\widetilde{P}_{6,k} + \widetilde{R}_{\mathrm{nav},k})K_k^{\mathsf{T}}, \quad H = s_k s_k^{\mathsf{T}} + C_{\mathrm{cmd}}, \\ C_{c,k} &= C_{\mathrm{cmd}} + \sigma_\theta^2 \operatorname{tr}(H) I_3 + (\sigma_{\mathrm{mag}}^2 - \sigma_\theta^2)H. \end{aligned} \tag{7.6}$$

The nested transform costs 51 outer points times 6 inner points, or 306 propagations per arc (some have zero weight). Across 216 arcs that is 66096 point propagations per objective/constraint evaluation before derivatives. Batch by phase and substep count, reuse compiled local maps, and preserve the sparse Jacobian assembly. This is a cost estimate, not a measured runtime. Four is a starting fidelity, not an accuracy guarantee; increasing it also increases the process-noise quadrature dimension.

8 Stochastic cost, constraints and decision variables

The existing Earth–Mars stochastic objective is a *hybrid robust cost*, not an energy integral:

$$J_{\text{old}} = \sum_k \ell_k, \qquad \ell_k = q_\eta(s_k) + \frac{1}{9}\,\text{tr}(\widetilde{P}_{6,k}) + 1000\,\text{tr}(C_{c,k}). \tag{8.1}$$

Its constructor explicitly rejects unequal arc durations. Simply removing that guard would give short planet-centred arcs the same cost weight as multi-week cruise arcs. The proposed nonuniform extension is

$$\boxed{J_{\text{robust}} = \sum_k w_k \ell_k, \qquad w_k = \frac{N\Delta t_k}{t_f - t_0}, \qquad N = 216.} \tag{8.2}$$

It preserves the original objective exactly when all arcs have equal duration. Its weights depend on the free Mars exit time and must be differentiated. This is an explicit proposed extension, that will be implemented for the psyche case.

The existing executed-thrust chance approximation is

$$q_\eta(s_k) + \sqrt{\chi^2_{3,1-\epsilon_u}}\,\rho(C_{c,k}) \le 1, \quad \rho(C) = \sqrt{\lambda_{\max}(C)}, \quad \epsilon_u = 0.01. \tag{8.3}$$

The implementation uses its existing smooth analytic three-dimensional spectral-radius approximation (smoothing and radius floor 10^{-6}). Conditional Gates errors make the unconditional control distribution a mixture and it should not be used anymore in the control chance constraints. Therefore the Gaussian confidence interpretation is approximate and requires Monte Carlo validation. Per-arc 99% confidence does not imply 99% confidence for all 216 arcs jointly.

Do not Retain mean surface exclusion and the mass floor at the existing mesh/integrator samples, with no new SOI path inequalities. Mars gets the event safety constraint developed below.

The terminal covariance constraint acts on all six position/velocity channels and their cross-covariance:

$$P_{6,N} \preceq P_{f,\text{target}}, \qquad I - P_{f,\text{target}}^{-1/2} P_{6,N} P_{f,\text{target}}^{-1/2} = GG^{\mathsf{T}}. \tag{8.4}$$

Here G is lower triangular; the current diagonal floor 10^{-5} slightly restricts the boundary of this feasible set. A proposed inherited absolute benchmark is

$$P_{f,\text{target}} = \text{diag}[(2000\text{ km})^2 I_3, (0.002\text{ km/s})^2 I_3].$$

Reusing Earth–Mars's variance reductions 225 and 100 instead would give 133.333 km per position axis and velocity limits equal to the new marginal launch standard deviations divided by 10. Neither choice is supplied by a Psyche operations covariance requirement.

The NLP variables would be: seven-component means at the stitched nodes; three-component normalised feedforward controls per arc; 3×6 gains per arc; 28 lower-triangular covariance-factor entries at each noninitial node; 21 terminal-slack entries; one Mars exit-duration variable; and an auxiliary nominal closest-approach time if represented explicitly. Fix the initial covariance, the three nominal Cartesian boundary states and their epochs as agreed. Mass is continuous, initially fixed in mean, and free at the later fixed Cartesian events. Covariance and gains remain optimisable through Mars entry. Gain scaling, the scaled gain bound 20, and node covariance-factor scaling should be retained and reported.

Writing $\mathcal{M}_\mu, \mathcal{M}_P$ for the two outputs of Eq. (7.5), the stochastic shooting equalities are

$$\begin{aligned} &\bar{x}_{k+1} - \mathcal{M}_\mu(\bar{x}_k, P_k, s_k, K_k, \Delta t_k) = 0, \\ &P_{k+1} - \mathcal{M}_P(\bar{x}_k, P_k, s_k, K_k, \Delta t_k) = 0, \qquad \widetilde{P}_k = S_k L_k L_k^{\mathsf{T}} S_k^{\mathsf{T}}. \end{aligned} \tag{8.5}$$

Here S_k is the fixed node-conditioning scale obtained from the covariance seed, and L_k is the lower-triangular decision factor. Only the 28 independent covariance residuals are enforced. At phase links, compose the moment map with the deterministic frame/scale transformation. Seed-conditioning scales are not additional physical uncertainty.

Keep the present JAX/SNOPT settings initially: 50000 major and 5000000 minor iterations, major optimality tolerance 10^{-5}, major/minor feasibility tolerance 10^{-8}. Do not import the deterministic IPOPT tolerances as though they belonged to this script. Retain JAX 64-bit arithmetic and no arbitrary runtime limit. The initial reference/gain restoration may need physical residual scaling to meet the new phase and B-plane constraints; scaling changes must be documented.

9 Osculating B-plane with a variable incoming asymptote

Use Mars-centred J2000 r, v at closest approach. Vector formulas avoid classical-element singularities at zero inclination. They represent the same osculating Kepler hyperbola; no conversion to node/periapsis angles is needed. Following Farnocchia Eqs. (1)–(6), define

$$\begin{aligned}
&C = v^{\mathsf{T}} v - 2\mu_M / \|r\|, \qquad v_\infty = \sqrt{C}, \\
&h = r \times v, \quad \hat{h} = h/\|h\|, \quad e_v = (v \times h)/\mu_M - r/\|r\|, \quad e = \|e_v\|, \\
&\hat{P} = e_v/e, \quad \hat{Q} = \hat{h} \times \hat{P}, \quad \boxed{\hat{S} = \frac{\hat{P} + \sqrt{e^2-1}\hat{Q}}{e}}, \qquad (9.1) \\
&B = \frac{\hat{S} \times h}{v_\infty}, \quad \hat{T} = \frac{V \times \hat{S}}{\left\|V \times \hat{S}\right\|}, \quad \hat{R} = \hat{S} \times \hat{T}, \quad V = (0,0,-1)^{\mathsf{T}}, \\
&B_R = B^{\mathsf{T}} \hat{R} = \frac{\hat{T}^{\mathsf{T}} h}{v_\infty}, \qquad B_T = B^{\mathsf{T}} \hat{T} = -\frac{\hat{R}^{\mathsf{T}} h}{v_\infty}. \qquad (9.2)
\end{aligned}$$

Require $C > 0$, $e > 1$, $\|h\| > 0$ and $\left\|V \times \hat{S}\right\| > 0$. Monitor their margins rather than silently clipping a non-hyperbolic encounter into a hyperbolic one. Choose a different fixed reference axis before a solve if the last denominator is poorly conditioned. Do not switch axes discontinuously inside the optimiser.

There is a convention difference in the supplied papers, Use Farnocchia's convention consistently.

9.1 Scaled coordinates and the physical clearance radius

Choose $\rho = R_M + h_{\text{safe}}$ and define, following Farnocchia Eqs. (32)–(36),

$$\lambda_\rho = \sqrt{1 + \frac{2\mu_M}{\rho C}}, \qquad \boxed{b_\rho = \lambda_\rho^{-1} \begin{bmatrix} B_R \\ B_T \end{bmatrix}}. \qquad (9.3)$$

These scaled coordinates still have units of length. They are not dimensionless. If $\hat{b} = b_\rho/\rho$ is used inside the NLP, transform covariance as $C_{\hat{b}} = C_b/\rho^2$ and compare with unit radius.

At actual closest approach, $r^{\mathsf{T}} v = 0$. Writing $r_c = \|r\|$, an osculating hyperbolic state satisfies

$$\|B\|^2 = r_c^2 + \frac{2\mu_M r_c}{C}, \qquad \|b_\rho\|^2 - \rho^2 = \frac{(r_c - \rho)[C(r_c + \rho) + 2\mu_M]}{C + 2\mu_M/\rho}. \qquad (9.4)$$

Thus $r_c \geq \rho$ is equivalent to $\|b_\rho\| \geq \rho$ at that event. This algebraic identity does not require the *entire* trajectory to be Keplerian; Farnocchia Section 3 explains why evaluation at actual closest approach matters for non-Keplerian motion. The propagation to that event still uses the powered model.

10 Full B-plane Jacobian: do not freeze $\hat{\mathbf{S}}$ or the focusing

The following partial derivatives, adapted from Farnocchia et al., Eqs. (17)–(26) and (34)–(35), define the analytic Jacobian. Automatic differentiation should reproduce them. Here, $C = v_\infty^2$ denotes the characteristic energy (called α in Farnocchia), $e = \|\mathbf{e}_v\|$, and μ_M is the gravitational parameter of Mars. As in Farnocchia, ∂ denotes a generic partial derivative with respect to a Cartesian position or velocity component.

The partial derivatives of the characteristic energy and asymptotic speed are

$$\begin{aligned} \partial C &= 2\mathbf{v}\cdot\partial\mathbf{v} + \frac{2\mu_M}{\|\mathbf{r}\|^3}\mathbf{r}\cdot\partial\mathbf{r}, \\ \partial v_\infty &= \frac{\partial C}{2v_\infty}. \end{aligned} \tag{10.1}$$

The partial derivatives of the eccentricity vector and magnitude are

$$\begin{aligned} \partial\mathbf{e}_v &= \frac{\partial\mathbf{v}\times\mathbf{h} + \mathbf{v}\times\partial\mathbf{h}}{\mu_M} - \frac{\partial\mathbf{r}}{\|\mathbf{r}\|} + \frac{\mathbf{r}\cdot\partial\mathbf{r}}{\|\mathbf{r}\|^3}\mathbf{r}, \\ \partial e &= \frac{\mathbf{e}_v\cdot\partial\mathbf{e}_v}{e}. \end{aligned} \tag{10.2}$$

The specific angular momentum satisfies

$$\begin{aligned} \partial\mathbf{h} &= \partial\mathbf{r}\times\mathbf{v} + \mathbf{r}\times\partial\mathbf{v}, \\ \partial\|\mathbf{h}\| &= \frac{\mathbf{h}\cdot\partial\mathbf{h}}{\|\mathbf{h}\|}. \end{aligned} \tag{10.3}$$

The partial derivatives of the perifocal axes are

$$\begin{aligned} \partial\hat{\mathbf{P}} &= \frac{\partial\mathbf{e}_v}{e} - \frac{\partial e}{e^2}\mathbf{e}_v, \\ \partial\hat{\mathbf{Q}} &= \frac{\partial\mathbf{h}\times\hat{\mathbf{P}} + \mathbf{h}\times\partial\hat{\mathbf{P}} - \partial\|\mathbf{h}\|\,\hat{\mathbf{Q}}}{\|\mathbf{h}\|}. \end{aligned} \tag{10.4}$$

The partial derivatives of the incoming asymptote and B-plane axes are

$$\begin{aligned} \partial\hat{\mathbf{S}} &= -\frac{\partial e}{e^2}\hat{\mathbf{P}} + \frac{\partial\hat{\mathbf{P}}}{e} + \frac{\partial e}{e^2\sqrt{e^2-1}}\hat{\mathbf{Q}} + \frac{\sqrt{e^2-1}}{e}\partial\hat{\mathbf{Q}}, \\ \partial\hat{\mathbf{T}} &= \frac{\mathbf{V}\times\partial\hat{\mathbf{S}} - [\hat{\mathbf{T}}\cdot(\mathbf{V}\times\partial\hat{\mathbf{S}})]\hat{\mathbf{T}}}{\|\mathbf{V}\times\hat{\mathbf{S}}\|}, \\ \partial\hat{\mathbf{R}} &= \partial\hat{\mathbf{S}}\times\hat{\mathbf{T}} + \hat{\mathbf{S}}\times\partial\hat{\mathbf{T}}. \end{aligned} \tag{10.5}$$

Then the unscaled B-plane coordinate derivatives are

$$\begin{aligned} \partial B_R &= \frac{\hat{\mathbf{T}}\cdot\partial\mathbf{h} + \mathbf{h}\cdot\partial\hat{\mathbf{T}} - B_R\,\partial v_\infty}{v_\infty}, \\ \partial B_T &= -\frac{\hat{\mathbf{R}}\cdot\partial\mathbf{h} + \mathbf{h}\cdot\partial\hat{\mathbf{R}} + B_T\,\partial v_\infty}{v_\infty}. \end{aligned} \tag{10.6}$$

For the gravitational-focusing factor

$$\lambda_\rho = \sqrt{1 + \frac{2\mu_M}{\rho v_\infty^2}}, \tag{10.7}$$

its partial derivative, for constant ρ, is

$$\partial\lambda_\rho = -\frac{2\mu_M\,\partial v_\infty}{\lambda_\rho \rho v_\infty^3}. \tag{10.8}$$

Finally, for $b_R = B_R/\lambda_\rho$ and $b_T = B_T/\lambda_\rho$,

$$\begin{aligned}\partial b_R &= \frac{\partial B_R - b_R\,\partial\lambda_\rho}{\lambda_\rho},\\ \partial b_T &= \frac{\partial B_T - b_T\,\partial\lambda_\rho}{\lambda_\rho}.\end{aligned} \tag{10.9}$$

Here ρ is a selected constant. Form the Jacobian

$$J_b = \frac{\partial(b_R, b_T)}{\partial(\mathbf{r}, \mathbf{v})}. \tag{10.10}$$

There is no direct mass column, but mass uncertainty has already affected the position/velocity covariance through the dynamics.

11 Off-nominal closest-approach time and covariance

For the nominal Mars passage, solve the continuous event

$$g(x) = r^\mathsf{T} v = 0, \qquad q_g = v^\mathsf{T} v + r^\mathsf{T} a > 0, \quad a = -\mu_M r/\left\|r\right\|^3 + T_{\max} s/m. \tag{11.1}$$

Search for all relevant stationary points and compare distances and phase endpoints. Do not select a maximum, a previous periapsis, or the closest mesh node. A read-only DOP853 propagation within the parent four-phase energy trajectory's Mars arc 59 (zero-based) gives an illustrative closest-approach altitude of about 5066.50 km and $C_3 \simeq 19.71037$ km^2/s^2. The revised three-phase energy solution has not yet been solved; this only locates the parent nominal branch for planning; it is not a certified continuous stochastic safety result. The event lies about 15.87 s into an arc of 178.04 s, which makes control-node proximity worth checking.

At the nominal event let $z = (r^\mathsf{T}, v^\mathsf{T})^\mathsf{T}$, $f_z = (v^\mathsf{T}, a^\mathsf{T})^\mathsf{T}$, and $n = (v^\mathsf{T}, r^\mathsf{T})^\mathsf{T}$. Farnocchia Eqs. (27)–(30) give

$$\boxed{\delta\tau = -\frac{v^\mathsf{T}\delta r + r^\mathsf{T}\delta v}{v^\mathsf{T} v + r^\mathsf{T} a}} = h_\tau \delta z, \qquad h_\tau = -n^\mathsf{T}/q_g, \tag{11.2}$$

$$\delta z_{\mathrm{CA}} = \delta z + f_z \delta\tau = E_{\mathrm{CA}}\delta z, \qquad E_{\mathrm{CA}} = I_6 - f_z n^\mathsf{T}/q_g. \tag{11.3}$$

Use total nominal *relative* acceleration, including thrust. A heliocentric implementation must first form consistent relative acceleration, including the moving origin's acceleration; the native Mars phase avoids that cancellation. White acceleration is not inserted as an instantaneous random value into a.

If $P_z(\bar\tau)$ is the covariance at the common nominal epoch, the intended mapping is

$$\boxed{J_{\mathrm{event}} = J_b E_{\mathrm{CA}}, \qquad C_b = J_{\mathrm{event}} P_z(\bar\tau) J_{\mathrm{event}}^\mathsf{T}.} \tag{11.4}$$

The joint B-plane/time covariance is

$$\mathrm{Cov}\begin{bmatrix}\delta b\\ \delta\tau\end{bmatrix} = \begin{bmatrix}J_{\mathrm{event}}\\ h_\tau\end{bmatrix} P_z(\bar\tau) \begin{bmatrix}J_{\mathrm{event}}\\ h_\tau\end{bmatrix}^\mathsf{T}. \tag{11.5}$$

This reports the time variance and its correlation with miss distance. E_{CA} is rank deficient because it projects onto an event surface; do not force its six-dimensional covariance to have a nonsingular Cholesky factor. Map directly to the two-dimensional C_b.

If covariance is available at an earlier epoch, first propagate it to $\bar{\tau}$, including process and control errors.

11.1 Explicit closest-approach shooting node

The nominal closest approach is represented by an explicit shooting node whose epoch is an optimisation variable. Using control points at events such as periapsis or flyby closest approach is a recognised multiple-shooting approach; see NASA's trajectory-reconstruction description. Locating the nominal closest approach and accounting for uncertain closest-approach times remain separate tasks.

The Mars segment is divided into two numerical segments:

$$t_{\mathrm{Mars,in}} \xrightarrow{\text{Mars dynamics}} \tau_c \xrightarrow{\text{same Mars dynamics}} t_{\mathrm{Mars,out}}. \tag{11.6}$$

Introduce the nominal closest-approach epoch, state and covariance:

$$\tau_c, \qquad \bar{\mathbf{x}}_c = \begin{bmatrix} \bar{\mathbf{r}}_c \\ \bar{\mathbf{v}}_c \\ \bar{m}_c \end{bmatrix}, \qquad P_c. \tag{11.7}$$

Here, τ_c is the free nominal closest-approach epoch, $\bar{\mathbf{x}}_c$ is an ordinary optimised shooting-node state, and P_c is the state covariance at that common nominal epoch. Neither the closest-approach state nor its epoch is fixed to an ephemeris value. They are determined by the dynamics, optimisation objective and constraints.

At the new node, impose

$$\boxed{\bar{\mathbf{r}}_c^{\mathsf{T}} \bar{\mathbf{v}}_c = 0.} \tag{11.8}$$

This equality makes the nominal radial velocity zero. The minimum-selection condition is

$$\boxed{q_c = \|\bar{\mathbf{v}}_c\|^2 + \bar{\mathbf{r}}_c^{\mathsf{T}} \bar{\mathbf{a}}_c > 0.} \tag{11.9}$$

The acceleration $\bar{\mathbf{a}}_c$ must be consistent with the nominal trajectory representation and include thrust and the adopted gravity model. The numerical implementation must handle this positivity condition explicitly: the stationarity equality alone also admits other stationary distances.

The state and covariance on either side of the new node remain linked by the usual shooting equations. There is no physical reset or change of gravity model at this node.

Meshes tied to the closest-approach epoch. To avoid tracking a moving event bracket, let the inbound and outbound Mars meshes follow τ_c. Define $t_{\mathrm{in}} = t_{\mathrm{Mars,in}}$ and $t_{\mathrm{out}} = t_{\mathrm{Mars,out}}$, and choose fixed normalised node fractions s_k^- and s_k^+ for the two segments. Their physical node epochs are

$$t_k^- = t_{\mathrm{in}} + s_k^- (\tau_c - t_{\mathrm{in}}), \tag{11.10}$$

$$t_k^+ = \tau_c + s_k^+ (t_{\mathrm{out}} - \tau_c), \tag{11.11}$$

with

$$t_{\mathrm{in}} < \tau_c < t_{\mathrm{out}}. \tag{11.12}$$

The normalised fractions remain fixed while the segment durations change. The optimiser can therefore move the nominal closest approach while retaining a fixed mesh topology.

Control treatment at the inserted node. Controls are held constant over each control arc. Introducing a new control update exactly at closest approach could make the acceleration discontinuous there. Samples reaching closest approach earlier or later could then encounter different controls, complicating the single-Jacobian timing approximation.

The inserted node is therefore initially treated as a shooting/evaluation node without an additional control update:

- Continue the same held executed command through the split for each stochastic realisation.
- Do not redraw navigation or execution errors solely because the numerical node has been added.

12 Probabilistic Mars clearance constraint

The physical requirement for the controlled passage is

$$\mathbb{P}\left[\min_{t\in[t_{M,\mathrm{in}},t_{M,\mathrm{out}}]} \|r_M(t)\| \geq \rho\right] \geq 1-\epsilon_M. \tag{12.1}$$

For a unique, well-resolved hyperbolic closest approach and small enough dispersion, approximate the event B-plane distribution by $b \sim \mathcal{N}(\bar{b}, C_b)$ using Eq. (11.4). Adapt Anibha Eq. (55) and Nuccio Eq. (14) as

$$\boxed{\left\|\bar{b}\right\| - \sqrt{\chi^2_{2,1-\epsilon_M}}\,\sqrt{\lambda_{\max}(C_b)} \geq \rho.} \tag{12.2}$$

For the selected $\epsilon_M = 0.001$, the two-dimensional multiplier is $\sqrt{\chi^2_{2,0.999}} = 3.716922$, not the one-dimensional 99.9% value 3.090232. The encounter safety target is 99.9%; the separate per-arc thrust risk remains 0.01.

For a smooth conservative two-dimensional numerical version, scale lengths by ρ, write $C_{\hat{b}} = \left[\begin{smallmatrix} a & c \\ c & d \end{smallmatrix}\right]$, and use

$$\bar{\lambda}_{\max} = \frac{a+d}{2} + \sqrt{\left(\frac{a-d}{2}\right)^2 + c^2 + \delta_\lambda^2}, \quad \bar{\rho}_b = \sqrt{\bar{\lambda}_{\max}}. \tag{12.3}$$

This is an upper bound to the exact standard-deviation radius. Start with dimensionless $\delta_\lambda = 10^{-6}$ (covariance scale) report their effect on clearance; these are proposed numerical parameters, not values from the papers. At zero covariance the resulting margin is of order tens of metres for Mars, so it is not literally zero. Reduce them if that artificial margin is material. Do not apply an upper smoothing to $\left\|\bar{b}\right\|$ on the left, which could relax safety. Away from $\bar{b} = 0$ its exact norm is smooth. Apply this hyperbolic B-plane construction to *Mars*. It is not a suitable collision constraint at the final asteroid-centre rendezvous, where relative velocity is zero and asteroid gravity is absent. The Earth departure is outside the revised optimisation interval.

13 Optional passive-safety path constraint

Nuccio Section IV.A.5 and Eq. (14) concern safety after complete loss of maneuverability: propagate ballistically from selected loss-of-control nodes to the next closest approach. This is stronger than checking the actual controlled passage and can change feasibility.

We will test this passive-safety path constraints on initial discretization node of the discretization arcs included in the time interval from 10 days before mars soi entrance until closest passafe, but not immediately only when i confirm and give the go for the implementation. For each selected interval define a separate failure flow with $T = 0$, no feedback and no execution error thereafter,

while retaining any agreed environmental acceleration diffusion. Map its state/covariance to its own future closest approach and impose

$$\left\|\bar{b}_j^{\text{loss}}\right\| - \sqrt{\chi^2_{2,1-\epsilon_j}}\sqrt{\lambda_{\max}(C_{b,j}^{\text{loss}})} \geq \rho. \tag{13.1}$$

At deterministic zero thrust use exactly zero physical mass flow, rather than interpreting norm-smoothing fuel loss as a real failed thruster. The inherited smoothed stochastic mass law remains the nominal operational model.

The requirement is conditional safety for *each separately postulated failure time*. This optional family is not part of the recommended first controlled-flyby benchmark unless approved.

14 Validation plan and implementation boundaries

After approval, implement the Psyche case through a small multiphase adapter and reusable geometry/event modules, rather than copying the entire stochastic solver. The existing single-phase cases should retain their defaults. The following checks are needed before interpreting convergence as robust mission performance:

1. **Reference fidelity.** Verify the manifest hashes, all 216 mesh intervals, native SI states, fixed-event states, ephemeris coverage and origin transforms. Restore the energy solution and report changes caused by RK9 and smoothed mass flow separately from stochastic changes.

2. **Initial distribution.** Verify the Gaussian 3σ covariance in Eq. (5.2), the 2000 km position standard deviations, and the velocity covariance in Eq. (**??**). Use the SPICE mean at the fixed exit epoch and the covariance in Eq. (5.5).

3. **Derivatives.** Compare analytic and automatic B-plane Jacobians with directional finite differences; verify moving-axis and focusing terms, the pure-Kepler cancellation, and event derivatives by independently relocated perturbed events. Check duration and frame derivatives and the control-covariance identity (7.6). When third-body gravity is added, evaluate Δf at both Mars SOI boundaries using the actual spline derivatives, and quantify the effect of dispersed crossing times before describing that residual as negligible.

4. **Integration and moments.** Compare four and eight RK9 substeps on every phase, then increase as needed. Set the Monte Carlo override to at least four as well; its current shared default of three must not carry into Psyche. Check states, covariances, gains, chance margins and derivative changes in both physical and NLP-scaled units. Retain the 216 arcs; increase integration fidelity rather than silently adding arcs. RK9 order for the smooth drift does not imply ninth-order accuracy for the SDE or its moments.

5. **Continuous Monte Carlo.** Use the agreed initial distribution and inherited OD, acceleration and execution errors. Carry each sample continuously through phase translations without resetting at Mars entry or arrival. Recompute its commanded thrust at control nodes, hold it over arcs, and find its own minimum distance. Use finer coupled Wiener paths to check event/altitude stability. Report invalid hyperbolic geometry rather than discarding unsafe samples.

6. **Probability reporting.** Begin with the existing 50000 samples and seed 42, add independent seeds, .

7. **Outputs.** Report solver status, scaled/physical residuals, energy and propellant, robust objective components, covariance and thrust histories, the variable Mars exit, closest-approach epoch/altitude, B-plane confidence plots and event-time histograms. Reuse the existing plotting style. Keep actual-flow and failure-flow safety results separate.

The saved deterministic local defect tolerance 5×10^{-14} is not a bound on global physical accuracy, stochastic quadrature error or uncertainty-model validity. Do not advertise an equivalent accuracy merely by retaining that number. Four RK9 substeps satisfy the requested minimum; adequate accuracy and convergence remain to be demonstrated. No stochastic Psyche run, gain computation, or solver implementation has been performed in preparing this document.

References

[1] D. Farnocchia, S. Eggl, P. W. Chodas, J. D. Giorgini and S. R. Chesley, "Planetary encounter analysis on the B-plane: a comprehensive formulation," *Celestial Mechanics and Dynamical Astronomy*, 131, article 36, 2019. doi:10.1007/s10569-019-9914-4. Supplied PDF: `s10569-019-9914-4.pdf`.

[2] A. A. Anibha, K. Oguri and J. Villa, "Autonomous Optical Navigation for Unknown Asteroids with Irregular Horizons," AAS 25-905, AAS/AIAA Astrodynamics Specialist Conference, Boston, 2025. Supplied short paper, especially pp. 11–13. Official conference programme. No DOI is assumed.

[3] G. Nuccio, K. Oguri and F. Topputo, "Statistical Trajectory Optimization Under Uncertainty for Icy Moon Exploration with Probability of Impact Constraints," 30th International Symposium on Space Flight Dynamics, Toulouse, 1–5 June 2026. Supplied PDF: `NUCCG01-26(5).pdf`. Politecnico di Milano publication record. The supplied conference version is the equation-numbering authority here.

[4] L. Messina, *GPU-Accelerated Robust Trajectory Optimisation for Low-Thrust Space Missions*, supplied MSc thesis manuscript, Delft University of Technology, 2026. Repository copy: `data/psyche_sources/MSc_Thesis_Lukas_Messina_Final_Version.pdf`. The attached "(9)" copy is byte-identical. Unfinished requirements are not treated as mission data.

[5] L. Messina's repository, inspected 25 October 2026: `stochastic_traj_opt_jax.py`, `xmds2_rk9.py`, the deterministic Psyche energy output, and `psyche_mission_events_reconstruction/README.md`. Exact inspected hashes are in the accompanying manifest.

[6] Tudat documentation, "Third-body accelerations." Official differential-acceleration formulation, accessed 25 October 2026.